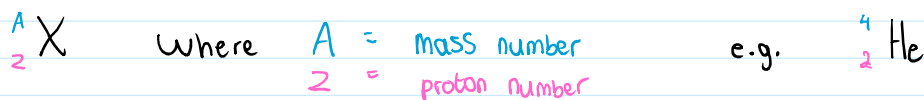
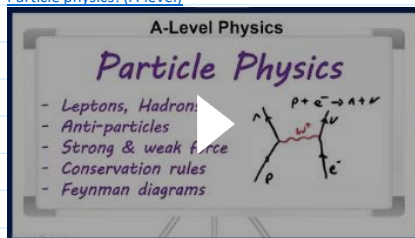
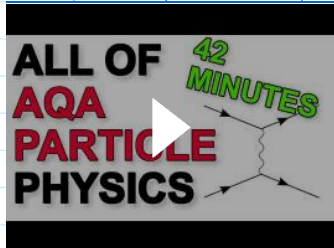


AQA A-Level Physics: Particles Topic

ALL of AQA Particle Physics in 42 minutes | A Level Physics Revision

Particle physics! (A-level)



Isotope: Same no. of protons, however different no. of neutrons
 Radioactive isotopes can be used to determine the **age of objects:**

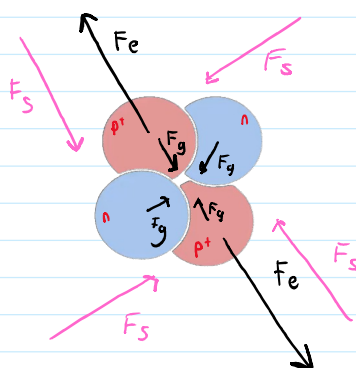
- Carbon 14 is radioactive and is breathed in from the atmosphere
- Once they die, the carbon 14 is trapped inside the object and will decay over time.
- We can then calculate the age of the object by using the percentage of Carbon 14 left inside, and comparing it with isotopic data.

forgot to say:
 All living beings have the same ratio of C-14 to C-12, therefore ...

Specific Charge = $\frac{\text{Charge}}{\text{Mass}}$ (C kg^{-1})
 per unit mass

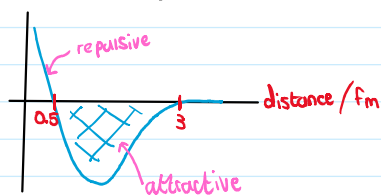
e.g. a proton would be
 $= \frac{1.60 \times 10^{-19} \text{ C}}{1.673 \times 10^{-27} \text{ kg}} = 9.56 \times 10^7 \text{ C kg}^{-1}$

Strong Nuclear Force:



Legend:
 F_g = Gravitational Force
 F_e = Electrostatic Force
 F_s = Strong Force

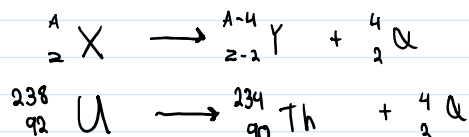
The Strong interaction is an attractive force that holds the nucleus together; it is the strongest fundamental interaction



$1 \text{ fm} = 1.0 \times 10^{-15} \text{ m}$ btw...

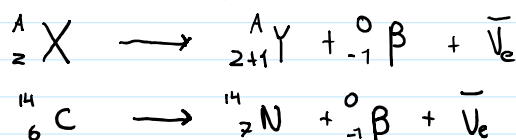
Below 0.5 fm , the Strong interaction is repulsive.
 Above 0.5 fm , the Strong interaction is attractive.
 The Strong interaction stops beyond 3 fm .

Alpha Decay



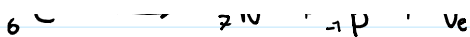
Alpha decay "removes" a helium nucleus from the atom, meaning the atom loses two protons and two neutrons.
 It releases an alpha particle, denoted ${}^4_2 \alpha$

Beta - Minus Decay



↑ this could

Beta-minus decay is completed the **Weak** interaction, allowing a neutron to turn into a proton.
 An anti-electron neutrino is also emitted as experiments showed us that the Beta particle,

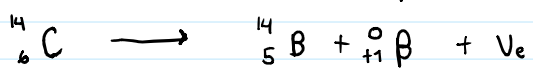
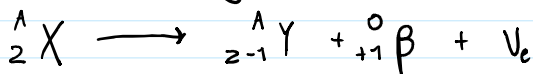


↑ You could also use ${}^0_{-1}\text{e}$, which is an electron

into a proton.

An anti-electron neutrino is also emitted as experiments showed us that the Beta particle, denoted ${}^0_1\beta$, as the total mass energy wasn't equal either side, so the $\bar{\text{Ve}}$ was introduced.

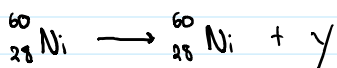
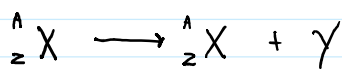
Beta-Plus Decay



↑ You could also use ${}^0_{+1}\text{e}$, which is a positron.

Similarly to Beta-Minus decay, Beta-Plus decay is also completed under the weak interaction; This allows a proton to become a neutron.

Gamma Decay



↑ Photon of high excess energy

An excited nucleus will eventually transition from a higher energy state to a lower energy state, releasing a high energy photon in the process.

That high energy photon ensures the conservation of energy law is followed.

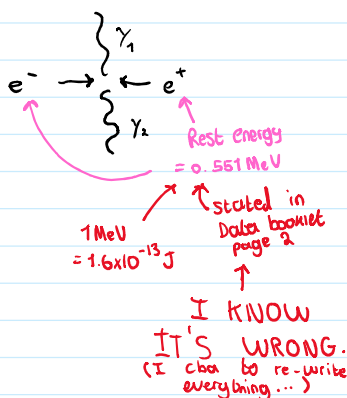
Antiparticles are particles of same mass however opposite charge, baryon number, and lepton number.

↳ are the counterpart to subatomic particles

The **energy of a photon** can be calculated using either $E = h \frac{c}{\lambda}$ or $E = hf$

Annihilation

When a particle meets its antiparticle, they collide and their masses get converted into energy.



The energy of each photon will be equal to that of the rest energy of both particles, divide by 2.

$$\therefore E_\gamma = \frac{0.511 + 0.511}{2} = 0.511 \text{ MeV}$$

Photon Frequency will be $E_\gamma = hf$ rearranged.

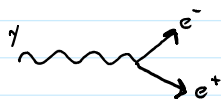
$$\therefore \frac{(0.511 \cdot 1.6 \times 10^{-13})}{6.63 \times 10^{-34}} = 8.31 \times 10^{19} \text{ Hz}$$

↑ energy in Joules

↑ Planck's Constant is given in Data Booklet page 1

Pair Production

The opposite of Annihilation. When a photon is converted into a particle and its antiparticle.



This photon must be above the threshold energy, which is the rest energy of both particles, in Joules.

$$\therefore (0.511 + 0.511) \cdot 1.6 \times 10^{-13} = 1.64 \times 10^{-13} \text{ J}$$

↑ correct value for e^- & e^+

↑ MeV to J

The photon's frequency is calculated the same way as above.

for e^- & e^+ 11eV to J

The photon's frequency is calculated the same way as above.

Mesons, Baryons, Leptons, and Quarks

Quarks are the fundamental particles. At A-Level, we know of the up, down, and strange quarks, where:

Type	Charge	Baryon number	Strangeness
u	$+\frac{2}{3}e$	$+\frac{1}{3}$	0
d	$-\frac{1}{3}e$	$+\frac{1}{3}$	0
s	$-\frac{1}{3}e$	$+\frac{1}{3}$	-1

Additional Note: Hadrons are Mesons and Baryons. (umbrella term)

Baryons are particles made up of 3 quarks.

For example, a proton is made up of two ups and a down



The two up quarks give a charge of $+\frac{4}{3}e$ ($\frac{2}{3} + \frac{2}{3}$)

The down quark gives a charge of $-\frac{1}{3}e$

This results in an overall charge of $+1e$ ($\frac{2}{3} + \frac{2}{3} - \frac{1}{3}$)

Furthermore, the Baryon number tells us if its a Baryon or not.

If its a baryon, $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

If not, $\frac{1}{3} - \frac{1}{3} = 0$ or $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$

Or, if its an antibaryon, $-\frac{1}{3} - \frac{1}{3} - \frac{1}{3} = -1$

Leptons are fundamental particles that are made up of nothing else.

Similar to a baryon, leptons have a lepton number.

+1 lepton number = lepton

-1 lepton number = antilepton

anything else = not a lepton

	Lepton number
Particles: $e^-, \nu_e; \mu^-, \nu_\mu$	+1
Antiparticles: $e^+, \bar{\nu}_e; \mu^+, \bar{\nu}_\mu$	-1

Mesons are particles made up of a quark and antiquark pair.

At A-Level, we only know of Kaons and Pions

π^0 u $\bar{u}$ d $\bar{d}$	π^- d $\bar{u}$	π^+ u $\bar{d}$	Pions $\rightarrow S=0$
K^0 d $\bar{s}$	K^- s $\bar{u}$	K^+ u $\bar{s}$	Kaons $\rightarrow S=+1$ or -1

Laws of Conservation

In every interaction, the following must be conserved:

- Charge
- Lepton Number
- Baryon Number
- Energy or Mass
- Strangeness (only under Strong interaction)

For example,

$$\begin{aligned}
 p &\rightarrow n + e^+ + \bar{\nu} \\
 \textcircled{A} \quad +1 &\rightarrow 0 + 1 + 0 \quad \checkmark \\
 \textcircled{B} \quad +1 &\rightarrow +1 + 0 + 0 \quad \checkmark \\
 \textcircled{C} \quad 0 &\rightarrow 0 - 1 - 1 \quad \times
 \end{aligned}$$

This decay is not possible as the lepton number is not conserved. ($0 \neq -2$, where LHS $\neq$ RHS)

For example,

$$\begin{aligned}
 \mu^- &\rightarrow e^- + \bar{\nu}_e + \nu_\mu \\
 \textcircled{A} \quad -1 &\rightarrow -1 + 0 + 0 \quad \checkmark \\
 \textcircled{B} \quad 0 &\rightarrow 0 + 0 + 0 \quad \checkmark \\
 \textcircled{C} \quad +1 &\rightarrow +1 - 1 + 1 \quad \checkmark
 \end{aligned}$$

This decay is possible.

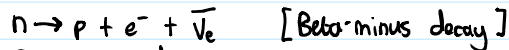
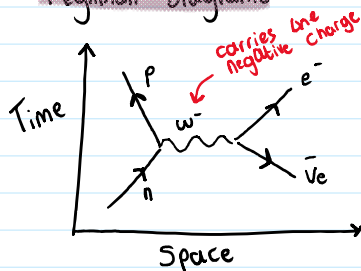
$$\begin{aligned} \textcircled{W} \quad 0 &\rightarrow 0 + 0 + 0 \quad \checkmark \\ \textcircled{L} \quad +1 &\rightarrow +1 - 1 + 1 \quad \checkmark \end{aligned}$$

Fundamental Interactions

- Electromagnetic
 - > γ , photon [exchange particle]
- Gravitational
 - > Graviton? (maybe) [exchange particle]
 - > Very weak force
- Weak Force
 - > W^- , W^+ , or Z^0 bosons [exchange particle]
 - > Affects all particles
- Strong Force
 - > π / gluon [exchange particle]
 - > Affects ONLY hadrons

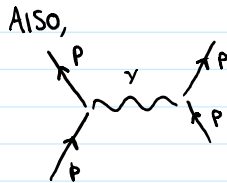
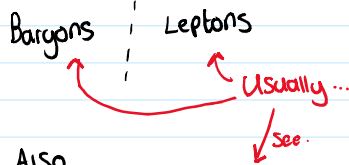
Exchange Particles are subatomic particles that mediate fundamental interactions between matter.

Feynman Diagrams



Beta-minus decay occurs under the weak interaction, so a W boson is the exchange particle.

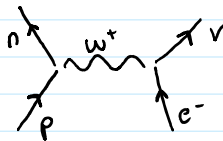
Since we are taking the negative charge from the neutron to the e^- , we use a W^- boson.



Electromagnetic repulsion of two protons.

γ (photon) as exchange particle since electromagnetic force.

Electron Capture



When a nucleus captures an electron from an inner-shell around the atom.

W^+ boson as exchange particle since positive charged moved away from the proton, and weak interaction.